

Week 7 Cost-Benefit Memo

AI-Assisted Emergency Department Triage

Vishal Baboolal | CariSurg MedTech Pathways Programme 2026

To: Dr. De Fretias, ED Board and Martina Griffith, Clinical IT Lead

Subject: Phase 3 model choice and the cost of added complexity

Verdict

Advance the optimised LightGBM model to a silent Phase 3 pilot, while retaining logistic regression as the transparent reference model and keeping both away from live clinical decisions.

Dataset and methods recap

The comparison used the same cleaned dataset and exact Week 6 80/20 stratified split with random_state=42. The dataset contains 55,121 emergency department encounters and 208 numeric clinical features.

Logistic regression and the bounded decision tree were retained as references. Random forest, LightGBM and XGBoost were screened as more sophisticated models. LightGBM had the strongest base accuracy among the complex candidates and was optimised using a training-only validation split.

Benchmark

Model	Accuracy	Macro F1	ESI 1	Train (s)	Infer (ms)	Explain
Logistic Regression	0.667	0.492	0.250	13.58	0.0103	High
Random Forest	0.497	0.398	0.500	7.30	0.0235	Medium
LightGBM Base	0.673	0.472	0.125	2.46	0.0325	Medium
XGBoost Base	0.614	0.377	0.188	3.57	0.0075	Medium
Optimised LightGBM	0.664	0.518	0.250	3.91	0.0574	Medium

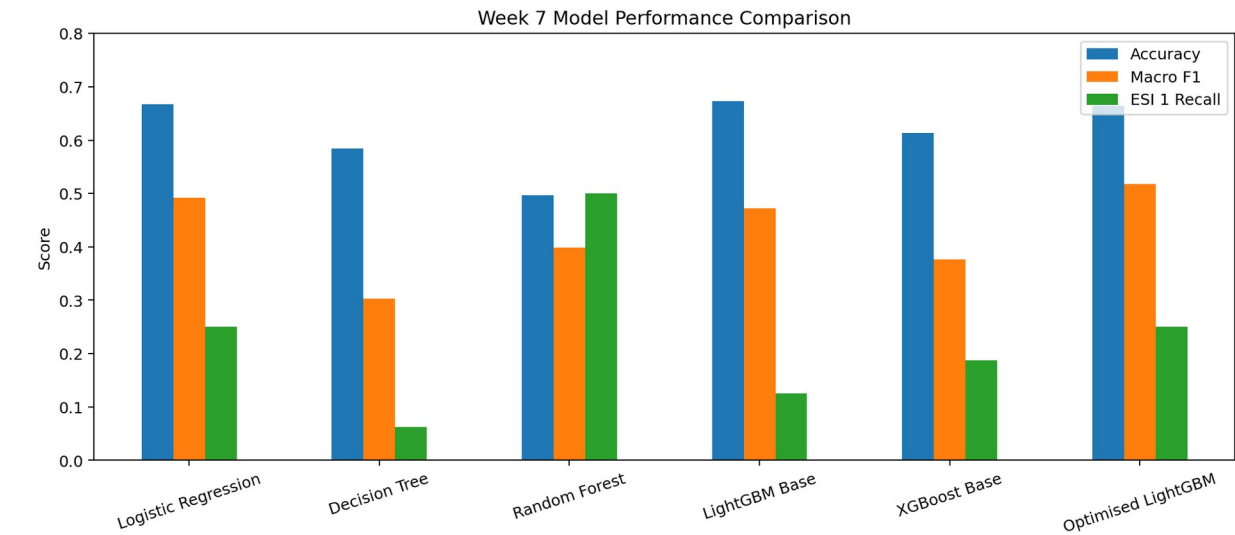


Figure 1. Performance comparison on the held-out test set.

Three arguments for advancing optimised LightGBM

- 1. Better balance across classes. Macro F1 increased from 0.492 for logistic regression to 0.518, while overall accuracy remained close.
- 2. Acceptable compute cost. The optimised model trained in 3.91 seconds and required about 0.0574 milliseconds per patient on the measured runtime.
- 3. A practical middle ground. LightGBM captures non-linear interactions and provides global feature importance, while remaining operationally lighter than a neural network.

Three arguments against advancing it

- 1. Reduced transparency. Logistic regression is easier to explain for one patient; a boosted ensemble combines many small trees.
- 2. Limited clinical gain. ESI Level 1 recall did not improve above logistic regression in this test set, despite the additional model complexity.
- 3. Additional maintenance. LightGBM adds another package, more hyperparameters, model-version controls and monitoring requirements.

Compute and deployability assessment

The measured differences are small relative to the duration of a clinical triage encounter. Inference remained below one millisecond per patient for every tested model. Nevertheless, these are wall-clock measurements from one runtime, not a Mercer capacity test.

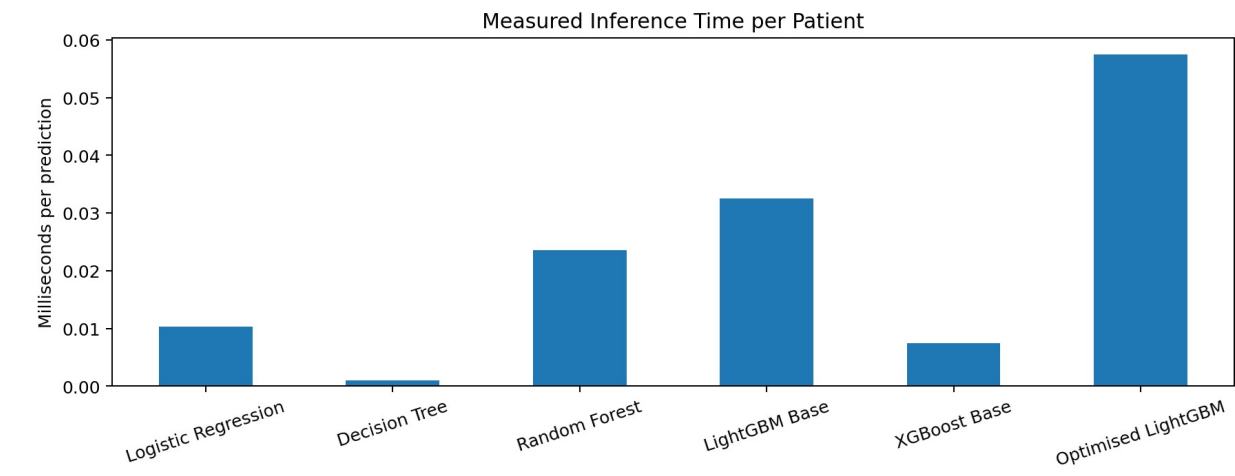


Figure 2. Measured inference time per patient.

Risks and unknowns

- The held-out test set contains only 16 ESI Level 1 cases, so critical-class performance remains statistically unstable.
- Subgroup fairness across age, sex, race, ethnicity, language, insurance status and arrival mode has not yet been established.
- Calibration, alert burden, data drift, missing-data behaviour and local Caribbean transferability remain untested.
- Global feature importance does not provide a complete patient-level explanation for one prediction.
- Training and inference times must be repeated on Mercer hardware and under realistic concurrent workload.

Cost-benefit judgement

The complex model buys a modest but measurable improvement in balanced performance at a negligible measured compute cost. The main price is reduced interpretability and additional governance work. That trade-off is justified for an offline silent pilot, but not for live clinical use.

Final recommendation

Advance optimised LightGBM to a silent Phase 3 pilot only.

Run it beside logistic regression without showing predictions to triage staff. Compare errors, subgroup performance, calibration and latency on local data. Do not allow either model to influence live prioritisation until the critical-class evidence is stronger and patient-level explanations have been reviewed with clinicians.

Single-line decision

Advance the optimised LightGBM model to a silent Phase 3 pilot, while retaining logistic regression as the transparent reference model and keeping both away from live clinical decisions.